

MODULE – I

1. Define Data Structure. List and explain the classification of data structure.
2. With examples explain pointer declaration, pointer initialization, and void pointer.
3. Explain the dynamic memory allocation functions in detail with examples.
4. Write the difference between Union and structure with an example. Also, explain the self-referential structure.
5. Define stack. Give the C implementation of push and pop functions. Include a check for empty and full conditions of a stack.
6. Problems related to stack.
7. Write the C function to convert an infix expression to a postfix expression.
8. Explain how a postfix expression is evaluated with suitable examples
9. What is polish/prefix notation? Explain with suitable examples.
10. Problems related to infix expression to a postfix.
11. Problems related to postfix/suffix evaluation.
12. Explain how to represent a polynomial using an array (Example). Write a C function to add two polynomials.
13. Define Sparse matrix. How to represent a sparse matrix in triple array form, explain with an example.
14. Write a C function to perform a simple transpose of a sparse matrix.
15. Write a C function for the Knuth, Morris, and Pratt pattern-matching algorithm. (See Problems also)

Module – II + III (1st Part)

1. Define queue. Explain the operation of a queue using arrays with suitable algorithms. (See problems related to queue).
2. Define circular queue. Explain the operation of a circular queue using arrays with suitable algorithms. (See problems related to circular queue).
3. What is traversing a linked list? Write a C function to print the elements of a linked list.
4. What is a singly linked list? Explain the algorithms for the insertion of a node in a singly

linked list with suitable C functions (beginning, middle, and end).

5. Explain the algorithms for the deletion of a node in a singly linked list with suitable C functions (beginning, middle, and end).

6. What is a stack? Explain the operation of a stack using linked stacks.

7. What is a queue? Explain the operation of a queue using a linked queue.

8. Explain doubly linked list. Write C functions to perform insertion and deletion at specific positions in a DLL.

9. What is a polynomial? Explain how a polynomial is represented using a linked List.

Write a C function to perform the addition of two polynomials using a circular linked List.

10. Write an algorithm to concatenate two lists.

11. What are the different types of linked lists? Write a C function to count the number of elements present in a single LinkedList.

12. Explain how to represent a sparse matrix using a Linked List with a proper example.

(Problems)

13. Write advantages of doubly linked list over singly linked list. Write a C function that will insert a given integer value into an ordered doubly linked list.

module 3,4 and 5

1. Write recursive C functions for inorder, preorder and postorder traversals of a binary tree.

2. Problems related to inorder, preorder, postorder traversals (...Given tree..).

3. Define Binary tree. Explain the representation of a binary tree with a suitable example.

4. Define the Threaded binary tree. Construct Threaded binary for the following elements: (...Given elements...).

5. Define complete, full, and skewed binary trees with proper examples.

6. Design an algorithm to traverse a graph using Depth First Search (DFS) and Breadth First Search (BFS). Apply DFS and BFS for the graphs given (...Given Graph.....).

7. Construct a binary tree from the Post-order and In-order sequence given. (From Pre-order and In-order)

8. Define Binary Search tree. Construct a binary search tree (BST) for the given elements:
(...Given Elements..) Traverse using in-order, pre-order, and post-order traversal techniques.
9. Define the Disjoint set. Consider the tree created by the weighted union function on the sequence of unions: union(0,1), union(2,3), union(4,5), union(6,7), union(0,2), union(4,6), and union(0,4). Process the simple find and collapsing find on eight finds and compare which find is efficient.
10. What is chained hashing? Discuss its pros and cons. Construct the hash table to insert the keys: (.....Given keys.....) in a chained hash table of 9 memory locations. Use $h(k) = k \bmod m$.
11. Define the leftist tree. Give its declaration in C. Check whether the given binary tree is a leftist tree or not (...Given tree...). Explain your answer.
12. What is dynamic hashing? Explain the following techniques with examples:
- i) Dynamic hashing using directories
 - ii) Directory less dynamic hashing
13. Define min Leftist tree. Meld the given min leftist trees. (...Given 2 trees....)
14. Define forest. Transform the given forest into a binary tree. (...Given forest..)
15. Show the adjacency matrix and adjacency list representation of the given graph.
(...Given graph...)
16. What is collision in hashing? What are the methods to resolve collision? Explain any one with proper example.
17. Explain optimal binary search tree with example.